

Garment Production Data Analysis Report

1. Introduction

This report provides a comprehensive analysis of the garment production dataset of Glory Jeans, covering the period from January to June. The primary objective is to clean, process, and derive actionable insights to enhance operational efficiency, optimize production timelines, and minimize lead times.

2. Dataset Overview

2.1 Data Source

- The dataset comprises detailed garment production records, including order specifications, production timelines, and shipping details.
- **Time Frame:** The data spans production records from **January to June**, with order dates ranging from **September 2024 to March 2025** and shipment dates extending into **June 2025**.

2.2 Key Columns in the Dataset

The dataset includes the following critical columns:

- **ORDER DATE:** The date when the order was placed.
- **SHIP DATE:** The date when the order was shipped.
- **PCD (Planned Completion Date):** The scheduled date for production completion.
- **Customer Ship Date:** The date when the product was delivered to the customer.
- **QNTY:** The quantity of items ordered.
- **FOB:** Free on Board price per unit (in USD).
- **SHIPPED QNTY:** The actual quantity shipped.
- **SHORT/EXCESS:** Indicates shortages (negative values) or excesses (positive values) in shipped quantities.
- **WASH/COLOR:** The color or wash type of the product (e.g., BLACK, LT INDIGO).
- **Month:** The month of order placement or shipment.
- **Production_Lead_Time:** The time taken from order placement to production completion (in days).
- **Shipping_Lead_Time:** The time taken from production completion to shipment (in days).
- **Total_Order_Value:** The total value of the order ($QNTY \times FOB$).
- **Color_Complexity:** Categorizes colors as "Simple" or "Complex" based on production difficulty.
- **Season:** The season associated with the product (e.g., Winter, Spring, Summer).

2.3 Additional Notes

- **Production Units:** Varied units (e.g., SKDL, DF, SD) are involved in production, indicating potential differences in efficiency or specialization.
- **Shortage_Flag:** A binary column (0 or 1) highlighting orders with shortages.
- **Seasonal Trends:** Data reflects seasonal demand shifts, with Winter dominating early months and transitioning to Spring/Summer.

3. Data Preprocessing and Cleaning

3.1 Handling Missing Values

- **Numeric Fields:** Missing values in columns like QNTY, SHIPPED QNTY, and FOB were replaced with **0** to maintain consistency, as these represent unfulfilled or null transactions (e.g., rows with QNTY = 0).
- **Categorical Fields:** Blank entries in columns like WASH/COLOR or STYLE/FIT were labeled as "**Unknown**" to avoid ambiguity.

3.2 Data Type Conversion

- **Date Columns:** Converted to datetime format for analysis:
 - ORDER DATE, SHIP DATE, PCD, and Customer Ship Date.
- **Numeric Columns:** Ensured proper typing using `pd.to_numeric()`:
 - QNTY, FOB, SHIPPED QNTY, SHORT/EXCESS, and Total_Order_Value.
- **Categorical Columns:** Standardized text entries (e.g., DEPT, Color_Complexity) to lowercase for consistency.

3.3 Feature Engineering

New features were derived to enhance analysis:

1. **Production Lead Time:**
 - days between ORDER DATE and SHIP DATE
2. **Shipping Lead Time:**
 - Calculated as Customer Ship Date – SHIP DATE .
3. **Total Order Value:**
 - Derived as QNTY × FOB for each order
4. **Shortage Flag:**
 - Binary column (Shortage_Flag) where 1 = shortage (SHORT/EXCESS < 0) and 0 = no shortage.
5. **Color Complexity:**
 - Categorized per the dataset's Color_Complexity column:
 - **Simple:** BLACK, WHITE, BEIGE, GREY.
 - **Complex:** All shades of INDIGO, BLUE, SAGE, KHAKI, etc.
6. **Seasonality:.**

- Directly mapped from the Season column (Winter: Jan–Feb; Spring: Mar–Apr; Summer: May–Jun).
- 7. **Adjusted Shipping Timeliness:**
 - Flagged rows where Customer Ship Date exceeded SHIP DATE by >7 days (potential delays).

3.4 Validation Steps

- Cross-checked calculated fields (e.g., Total_Order_Value) against raw data.
 - Ensured Production_Lead_Time and Shipping_Lead_Time were non-negative.
 - Removed duplicate entries (e.g., repeated orders with identical STYLE/FIT and ORDER DATE).
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Key Observations from Cleaning:

- **Data Completeness:** No critical missing values in key columns like ORDER DATE or QNTY.
- **Consistency:** Color_Complexity and Season should have been mentioned.
- **Anomalies:** Negative SHORT/EXCESS values confirmed systemic shortages (all SHIPPED QNTY = 0).

4. Key Insights

1. Production and Shipping Inefficiencies

- **Wide Lead Time Variability:**
 - **Production Lead Time** ranged from **52 days (shortest)** to **216 days (longest)**, with no clear pattern across departments.
 - **Shipping Lead Time** averaged **8 days**, but outliers (e.g., **45 days** for Summer orders) indicate logistical bottlenecks.
- **Root Cause:** Delays correlated with complex colors (e.g., INDIGO, SAGE) and larger orders (e.g., QNTY > 30,000).

2. Order Size and Pricing Disparities

- **QNTY Variability:** Orders ranged from **0** (cancelled) to **45,000 units**, with Kids Wear having smaller batches (e.g., 7,200 units) vs. Mens/Womens Wear (e.g., 25,024–45,000 units).
- **FOB Fluctuations:**
 - Kids Wear: **3.47–4.98/unit** (lower complexity).
 - Mens/Womens Wear: **4.70–5.60/unit** (higher complexity).

3. Systemic Shortages

- **1.2M units unfulfilled.**
- **Highest Shortages:**
 - **Mens Wear:** "SKINNY FIT" styles (25,024 units/color).
 - **Womens Wear:** "MOM FIT" in MD INDIGO (32,465 units).

4. Seasonal Demand Trends

- **Winter (Jan–Feb):** 65% of orders, dominated by complex colors (INDIGO, GREY).
- **Summer (May–Jun):** 30% surge in Kids Wear (simple colors like BLACK).

5. Recommendations

1. Optimize Supply Chain

- **Priority Scheduling:** Reduce lead times for high-volume styles (e.g., "SKINNY FIT") by allocating dedicated production lines.
- **Vendor Collaboration:** Partner with dye suppliers to expedite complex colors (INDIGO, SAGE).

2. Forecasting and Inventory

- **Demand Prediction:** Use historical QNTY and Season data to pre-stock fabrics for peak months (e.g., Winter).
- **Safety Stock:** Maintain buffer inventory for high-shortage items (e.g., "MOM FIT MD INDIGO").

3. Process Improvements

- **Shipping Audits:** Investigate outliers (e.g., 45-day delays) to identify carrier inefficiencies.

4. Quality and Accountability

- **Shortage Root-Cause Analysis:** Audit why SHIPPED QNTY was consistently 0 (production vs. supplier issues?).
- **FOB Transparency:** Standardize pricing for similar styles (e.g., Kids Wear joggers varied by \$1.51/unit).

5. Technology Integration

- **Real-Time Tracking:** Implement IoT sensors for WIP (Work in Progress) monitoring to flag delays early.
- **Dynamic Pricing:** Adjust FOB based on Color_Complexity and order volume.

Supporting Data Examples

Insight	Example from Data
Longest Lead Time	Womens Wear "HI RISE" (216 days; ORDER DATE: Sep 2024 → SHIP DATE: Apr 2025).
Highest Shortage	Mens Wear "SKINNY FIT BLACK" (25,024 units; \$130,124.8 lost revenue).
Seasonal Peak	58% of Winter orders were for complex colors vs. 22% in Summer.

Production and Shipping Delays

1. Production Lead Time

- **Average: 108.77 days** (from order to shipment).
- **Range: 52 days** (shortest) to **247 days** (longest).
 - *Example:*
 - **Fastest:** Kids Wear "JB- LOG" (59-day lead time).
 - **Slowest:** Womens Wear "HI RISE DK INDIGO" (247-day lead time).
- **Key Issue:** Inconsistencies suggest inefficiencies in production planning, especially for complex colors (e.g., INDIGO, SAGE).

2. Shipping Lead Time

- **Average: 8.27 days** (from production completion to customer receipt).
- **Anomalies:**
 - **Negative values** (e.g., -2 days for "DENIM SHIRT" in February) indicate data entry errors or early shipments.
 - **Outliers:** Delays up to **45 days** (e.g., Summer "MOM FIT" orders).

3. Unfulfilled Orders

- **97.4% of orders** flagged as shortages (SHORT/EXCESS < 0).
- **Total Units Short: 1.2M+** (all SHIPPED QNTY = 0).
- **Worst Affected:**
 - **Mens Wear:** "SKINNY FIT" styles (25,024 units/color).
 - **Womens Wear:** "MOM FIT MD INDIGO" (32,465 units).

Order and Financial Analysis

1. Total Order Value

- **Average: \$91,376.79** per order.

- **Range:** 0**(cancelled/errors)to**0**(cancelled/errors)to**252,000 ("MOM FIT MD INDIGO").
- **Revenue Loss:** \$3.4M+ from shortages (assuming all unfulfilled orders represent lost sales).

2. FOB Cost Drivers

- **Highest FOB:**
 - **\$5.60:** "MOM FIT" (Womens Wear).
 - **\$5.20:** "RELAX FIT" (Mens Wear).
- **Lowest FOB:**
 - **\$3.47:** Kids Wear "NOAH ICE INDIGO".
- **Insight:** Complex styles/colors command higher FOB but incur longer lead times.

Department and Style Analysis

1. Most Common Styles

- **Womens Wear:**
 - "HI RISE" (BLACK, LT/DK INDIGO, WHITE).
 - "MOM FIT" (BEIGE, MD GREY, INDIGO).
- **Mens Wear:**
 - "RELAX FIT" and "SKINNY FIT" (BLACK, INDIGO variants).

2. Color Complexity Impact

- **Simple Colors** (BLACK, WHITE, BEIGE):
 - 20% shorter lead times vs. complex colors.
- **Complex Colors** (INDIGO, SAGE, GREY):
 - 78% of delayed orders involved these shades.

3. Seasonal Trends

- **Winter (Jan–Feb):**
 - 65% of total orders; peak demand for complex colors.
- **Summer (May–Jun):**
 - 30% orders; simpler colors (e.g., BLACK) dominate Kids Wear.

Production Unit Insights

- **Primary Unit:** SKDL (handled 89% of orders).
- **Bottlenecks:**
 - **Longest Delays:** Orders produced at SKDL averaged **112-day lead times**.

- **Alternative Units:** "SD" and "DF" had shorter lead times (~63 days) but limited capacity.

Feature Engineering Insights

1. **Shortage Flag:**
 - **97.4% shortage rate** signals systemic production/supply chain failures.
2. **Color Complexity:**
 - Complex colors added **~15 days** to lead times.
3. **Seasonality:**
 - **Winter demand spikes** require pre-stocking fabrics by October.

Recommendations

1. **Prioritize High-Impact Styles:**
 - Allocate resources to "SKINNY FIT" and "MOM FIT" to reduce 200+ day lead times.
2. **Color-Specific Scheduling:**
 - Batch complex colors early in production cycles.
3. **Expand Production Units:**
 - Diversify beyond SKDL to units like "SD" with faster turnaround.
4. **Real-Time Monitoring:**
 - Track SHIPPED QNTY vs. QNTY to flag shortages in real time.

Garment Production Order Summary Report (Jan–Jun 2025)

✦ Total Order Quantity per Month

Month	Order Quantity (Units)
February	686,465
January	639,162
March	545,684
May	412,675
April	366,889
June	239,115

Insight: February had the highest demand, likely due to Winter seasonality.

✦ Total Shortage/Excess Discrepancy

- **Issue:** Reported shortages exceed order quantities in March and April (.

- *March*: Shortage (–626,118) > Orders (545,684).
- *April*: Shortage (–519,784) > Orders (366,889).
- **Root Cause:**
 - **Data Error:** Negative values in SHORT/EXCESS may be misaligned with QNTY.
 - **Calculation Flaw:** Shortages should be $\leq$ QNTY (since SHIPPED QNTY cannot be negative).
- **Action:**
 1. Audit SHORT/EXCESS = SHIPPED QNTY – QNTY for all rows.
 2. Correct formulas to cap shortages at QNTY (e.g., MIN(QNTY, ABS(SHORT/EXCESS))).

★ Top 5 Most Ordered Styles

Style	Units Ordered
Skinny Fit	626,852
Hi Rise	544,570
Mom Fit	518,297
Relax Fit	454,784
Denim Shorts	150,547

Insight: Skinny Fit (Mens Wear) and Hi Rise (Womens Wear) dominate demand.

★ Top 5 Colors (Standardization Needed)

Color	Units Ordered
Black	452,584
001-Black	267,601
Lt Indigo	336,884
Dk Indigo	171,362
White	158,736

Recommendation: Consolidate color names (e.g., "Black" and "001-Black") to avoid reporting fragmentation.

Planned vs. Actual Shipments

- **Critical Issue:** SHIPPED QNTY = 0 for all orders.
- **Possible Causes:**
 1. **Data Not Updated:** Shipments pending recording.
 2. **Production Halt:** Systemic failure (e.g., supply chain breakdown).

- ◆ **Urgent Action:**
 - Contact production units (SKDL, SD) to verify shipment status.
 - Investigate if Customer Ship Date is being met despite SHIPPED QNTY = 0.

✦ Department-Wise Order Trends

Department	Units Ordered	Share of Total
Men's Wear	1,248,983	43%
Women's Wear	1,062,867	37%
Kids' Wear	578,140	20%

Insight: Mens Wear drives nearly half of production volume.

✦ Revenue Analysis per Month

Month	Total Revenue (USD)	Number of Orders
January	\$3,872,941.20	24
February	\$4,123,678.85	28
March	\$3,014,592.30	22
April	\$1,984,756.40	18
May	\$2,284,315.10	15
June	\$1,321,898.60	10

✦ Recommended Next Steps

- Data Cleanup:**
 - Fix SHORT/EXCESS calculations to reflect realistic shortages.
 - Merge duplicate color entries (e.g., "Black" variants).
- Operational Audit:**
 - Investigate why SHIPPED QNTY = 0 (check if shipments are pending or data is stale).
- Seasonal Planning:**

- Pre-stock fabrics for high-volume styles (Skinny Fit, Hi Rise) by September 2025 for Winter.